

Is Everything Purple

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Human brain is a fascinating thing - for all the advancements that we have made in the last centuries, there's still very little we know about brain and particularly its ability to create reality. Focus on create - one might ask what do you mean when you say that the brain 'creates' reality? Well ~~the~~ to understand that, we would first have to talk colors.

Colors are another fascinating, simple yet complex concept. I was recently watching a YT video about why do we have color wheels instead of a color line? For ~~one~~ one, we often study about the electromagnetic spectrum - light's different wavelengths and there's a small slice of this spectrum that we can visually experience - termed "Visible spectrum" it spans from roughly 380 nm to 700 nm with violet at 380 and red at higher wavelength. You can essentially sweep across this spectrum to get any color - well nearly any color! Purple doesn't exist on the spectrum. It just doesn't exist in reality! It's an entirely "mental color", produced in your brain when it senses combination of blueviolet and red light. So in a way our brain takes the visible light spectrum and converts

it into a color wheel that we experience, with purple/magenta sitting b/w violet and red.

So in theory, if we have one sensor that is able to detect the wavelength of a light, it wouldn't be able to 'sense' purple if it is limited to sensing only one light beam at a time.

As surprising as this little scientific anecdote is, it raises an even bigger question — What if everything is Purple? I don't mean it in a cartoonish sense that everything is purple painted, rather it's a question of epistemology and ontology. { Ontology = What exists/is, epistemology = How do we know } • What are our brains construct as a perception of reality is surprisingly fallible. It is, after all, just a perception. How do we know that, like purple, so many things are mere mental constructs, and if mental construct is even a good representation of reality or not? Our senses are our inputs, our bridge, our portal to the external reality, they perceive physical signals — light, sound, chemicals etc and based on these inputs, they generate electrical signals (essentially a time-series like signals) and our brain actively processes these signals to construct reality. That's why I said 'It creates reality'.

I am not going to indulge in the ontological arguments that whether this "external reality" is real or not, ~~I'm~~ here I'm more concerned with the process by which this plausible external reality gets translated into the individual experience of reality.

Imagine a blind person — someone who was born completely blind. The said person has never seen colors and he asks you to explain what Green is like $\frac{?}{?}$, how would you do that? How could you do that at all? Perhaps tell him about the wavelength and scientology of colors, or maybe that it's "calm", "like a breeze"? And imagine you tell him everything that has ever been said or written about the color green, would he come to know the color? I think not, a child who just saw green for the first time in his life would know more about it than this blind man.

Another thing to consider is that it is entirely possible that your experience of green could be different from mine. We sense the light of same physical characteristic, but there's no scientific way to see if the experience of green created by this physical light is the same experience in your mind as mine. Hence this construction of reality is highly personalized, so I ~~called~~ called it "individual reality".

Under normal circumstances, there exists a correlation b/w the external reality and this individual reality. After all, every time your eyes sense a light of a particular spectrum, it produces a more or less similar individual reality of color. Everytime you hike up and go near a physical lake, your brain constructs an individual reality of lake. But its not difficult for the brain to brake down and construct individual reality that is disconnected from the physical reality. Travellers in desserts see lakes where none exists (a "purple" lake), you see and feel like falling in your dreams, drugs induce elements into this shaky individual reality and we have example of people with functioning eyes not able to experience one side or other, a gap in the construction of reality!

Alas! What shaky foundations we have got for individuals possessing consciousness, and if this very consciousness is constructed through the experience of individual reality, then shaky it would be too. Or perhaps the very dissociation of physical and individual reality, the very struggle b/w them, the mental process of continuously constructing reality is what gives us consciousness and makes us human. A purple reality producing a purple consciousness!